

PRACHI PATEL

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Education

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering (AI & ML)

2023 – 2027

CGPA: 8.39/10.0

Relevant Coursework

- Data Structures & Algorithms
- Probability & Statistics
- Database Management Systems
- Linear Algebra
- Machine Learning
- Deep Learning
- Artificial Intelligence
- Data Visualization
- Statistical Learning
- Natural Language Processing

Projects

DocuMind RAG | Python, FastAPI, LangChain, ChromaDB, Streamlit, Groq/Gemini API

May 2026

- Engineered a production-ready Retrieval-Augmented Generation (RAG) system over a domain-specific corpus, with a fully decoupled FastAPI backend and Streamlit frontend communicating via REST.
- Implemented **hybrid retrieval pipeline** combining BM25 keyword search with dense vector similarity (ChromaDB) and token-aware chunking, improving retrieval precision over naive vector-only search.
- Designed a multi-LLM provider layer supporting Groq and Gemini with dynamic model switching, async PDF ingestion, and a retrieval inspection module exposing retrieved chunks for full transparency.
- Built a hand-curated evaluation harness of **30 question-answer pairs** to evaluate retrieval accuracy and response faithfulness, identifying and resolving failure cases where vector-only search missed exact-keyword queries.
- Integrated confidence-based abstention to reduce hallucinated answers when retrieval confidence was low, and downloadable chat history with session-level reset and undo controls.

Text2Query | Python, FastAPI, SQLite, LLM APIs (SQLCoder/GPT), Pydantic

April 2026

- Developed an autonomous Text-to-SQL agent that converts natural language questions into validated, executable SQL queries against a **multi-table relational schema** with foreign-key relationships.
- Designed a schema analysis pipeline that scans tables, columns, and foreign-key relationships to generate semantic context, reducing irrelevant information passed to the language model.
- Architected a self-correction loop using SQLite's **EXPLAIN** for pre-execution syntax validation and automatic error-feedback retries (capped at **3 attempts**), improving SQL execution success from a single-shot baseline.
- Enforced read-only query guardrails blocking destructive SQL operations (**DROP, DELETE, UPDATE**) before execution, mitigating prompt-injection and data-safety risks in agentic systems.
- Evaluated the agent against a hand-built set of **25 natural language queries** spanning single-table lookups to nested aggregations, measuring execution accuracy before and after iterative self-correction.

Technical Skills

Programming Languages: Python, SQL, Bash

Machine Learning: Scikit-learn, Pandas, NumPy, Feature Engineering, Exploratory Data Analysis (EDA), Data Preprocessing, Model Evaluation

Generative AI & LLMs: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic Workflows, Self-Correction Loops

LLM APIs: Groq API, Google Gemini API, OpenAI API, Ollama (Local Inference)

Vector Search & Databases: ChromaDB, SQLite, PostgreSQL, BM25, Vector Similarity Search

Frameworks & Deployment: FastAPI, Streamlit, Pydantic, RESTful APIs, Docker

Developer Tools: Git, GitHub, Jupyter Notebook, VS Code, Postman

Leadership / Extracurricular

Gujarati Club, VIT

2023 – 2025

Coordinator

VIT

- Led one of VIT Bhopal's largest cultural communities by organizing Garba Night with **5,000+ participants**.
- Oversaw **150+ volunteers** across marketing, sponsorships, logistics, ticketing, and crowd management, ensuring seamless event execution.
- Spearheaded multiple cultural initiatives that enhanced student engagement, cross-functional collaboration, and campus participation.